

EXPERIMENT 2

MONOALPHABETIC SUBSTITUTION CIPHER

AIM

To implement the Monoalphabetic Substitution Cipher using Python.

ALGORITHM

1. Read the plaintext from the user.
2. Define a substitution key containing all 26 unique alphabet letters.
3. Replace each plaintext letter with its corresponding letter from the substitution key.
4. Display the encrypted (cipher) text.
5. For decryption, replace each cipher letter with its corresponding original alphabet letter.
6. Display the decrypted text.

CODE

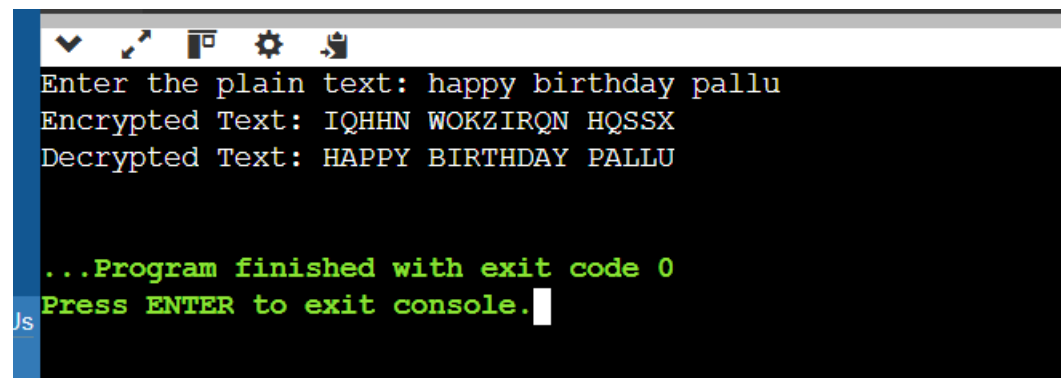
```
import string
alphabet = string.ascii_uppercase
key = "QWERTYUIOPASDFGHJKLZXCVBNM"
plain = input("Enter the plain text: ").upper()
# Encryption
cipher = ""
for ch in plain:
    if ch.isalpha():
        index = alphabet.index(ch)
        cipher += key[index]
    else:
        cipher += ch

print("Encrypted Text:", cipher)
```

```
# Decryption
decrypted = ""
for ch in cipher:
    if ch.isalpha():
        index = key.index(ch)
        decrypted += alphabet[index]
    else:
        decrypted += ch

print("Decrypted Text:", decrypted)
```

OUTPUT



```
Enter the plain text: happy birthday pallu
Encrypted Text: IQHHN WOKZIRQN HQSSX
Decrypted Text: HAPPY BIRTHDAY PALLU

...Program finished with exit code 0
Press ENTER to exit console.
```

RESULT

Thus, the Monoalphabetic Substitution Cipher was implemented successfully using Python.